

PrenticeMath Core Formula Sheet

Algebra & Core Math

Concept / Topic	Formula / Equation
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Slope Formula	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Slope-Intercept Form	$y = mx + b$
Logarithm Rule	$\log_b(xy) = \log_b(x) + \log_b(y)$

Geometry & Trigonometry

Shape / Concept	Formula / Identity
Area of a Circle	$A = \pi r^2$ (Circumference: $C = 2\pi r$)
Pythagorean Theorem	$a^2 + b^2 = c^2$
Trig Ratios (SOH CAH TOA)	$\sin(\theta) = \text{Opp/Hyp}$ $\cos(\theta) = \text{Adj/Hyp}$ $\tan(\theta) = \text{Opp/Adj}$
Pythagorean Identity	$\sin^2(\theta) + \cos^2(\theta) = 1$

Probability & Statistics

Concept	Formula / Description
Mean (Average)	$\mu = \frac{\sum x_i}{n}$
Sample Standard Deviation	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$
Probability of Event A	$P(A) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Outcomes}}$
Complement Rule	$P(A') = 1 - P(A)$
Addition Rule (General)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Permutations (Order matters)	$nPr = \frac{n!}{(n - r)!}$
Combinations (Order doesn't)	$nCr = \frac{n!}{r!(n - r)!}$